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Bitcoin Can Help Save The Planet

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Bitcoin | Electricity | Environment | Mining | Renewables | Sustainability

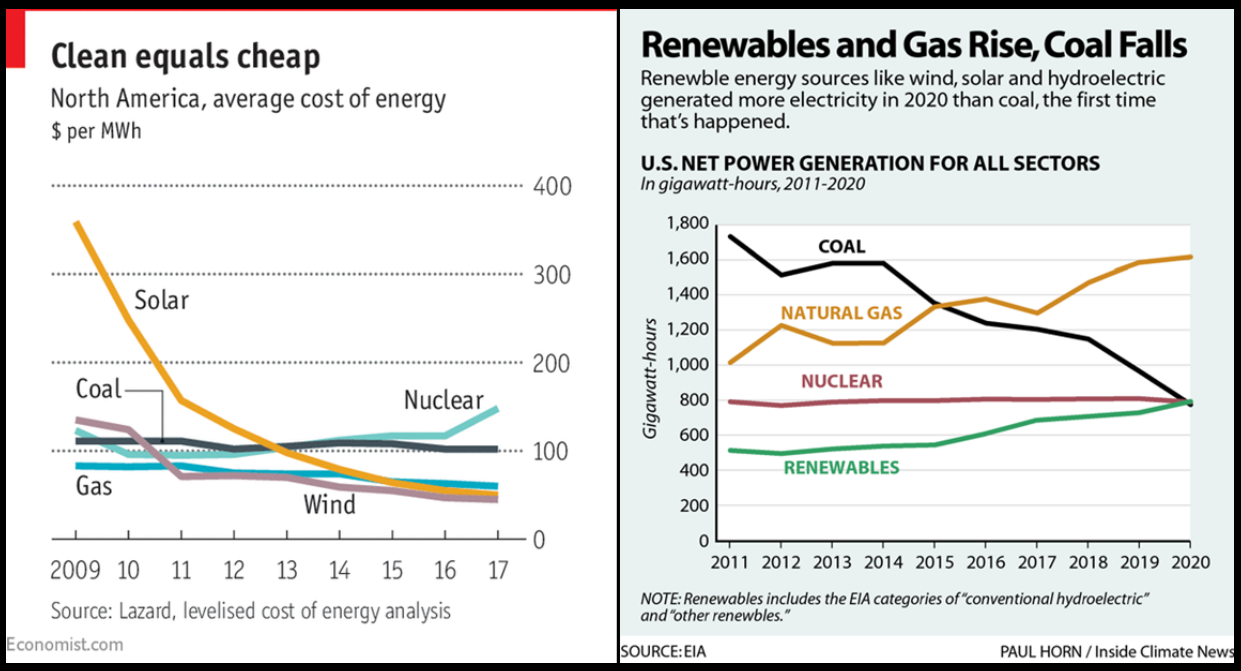
Disclosure: Delphi Ventures and members of our team hold positions in BTC. This statement is intended to disclose any conflict of interest and should not be misconstrued as a recommendation to purchase BTC. This is not investment advice.



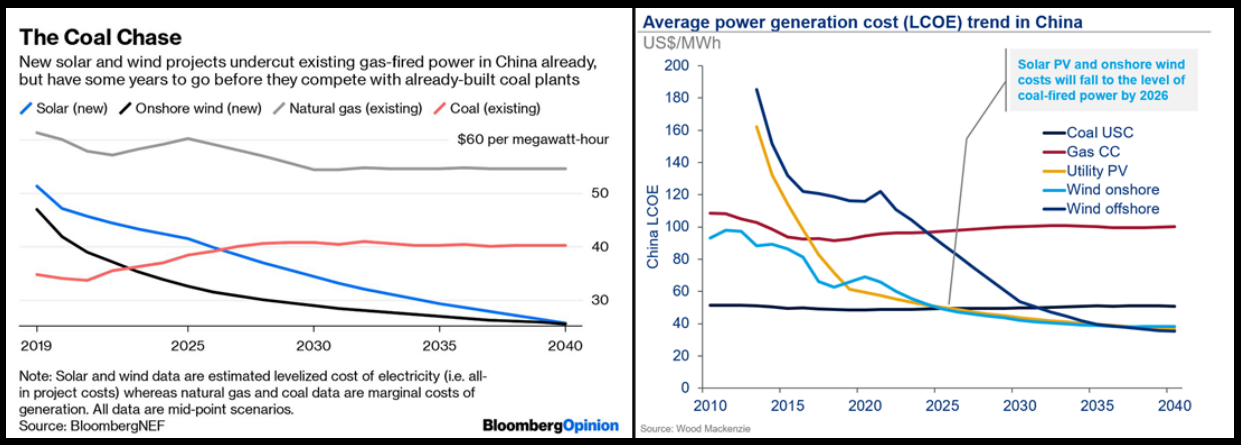
Recently, I was talking with a friend who works in ESG investing. For those unfamiliar with the acronym, ESG stands for "environmental, social and governance". As the name implies, the purpose of ESG is to evaluate investments by their environmental sustainability and societal impact. He had mentioned that institutional clients of his were becoming increasingly interested in Bitcoin as a potential investment, but they were concerned with the negative environmental impact from its high energy consumption. This is certainly a valid point because, frankly, Bitcoin consumes a lot of energy. As an individual who believes in climate change and the need for society to take action, I can understand why people form a negative opinion towards Bitcoin when they see a headline that says 'Bitcoin consumes more electricity than Argentina'. This may be true at face value but the point it's trying to convey is somewhat misleading. The situation is far more nuanced than the attention-grabbing headlines typically depict. In this post, I'll articulate the argument for why Bitcoin can actually have a positive impact on the environment.

Let's start by putting ourselves in the shoes of a Bitcoin miner to better understand the position they're in. As a miner you have one objective – be profitable. To accomplish this in such a hyper-competitive market, you need to reduce your costs as much as possible. Miners typically have two primary types

of capital expenditures – 1) mining hardware and 2) electricity. The hardware is the upfront expense, while electricity is your recurring expense. It's the electricity component that often dictates whether or not a miner is profitable or losing money. As a result, every miner around the world is looking to purchase the most energy efficient machines and source the cheapest electricity to run them.

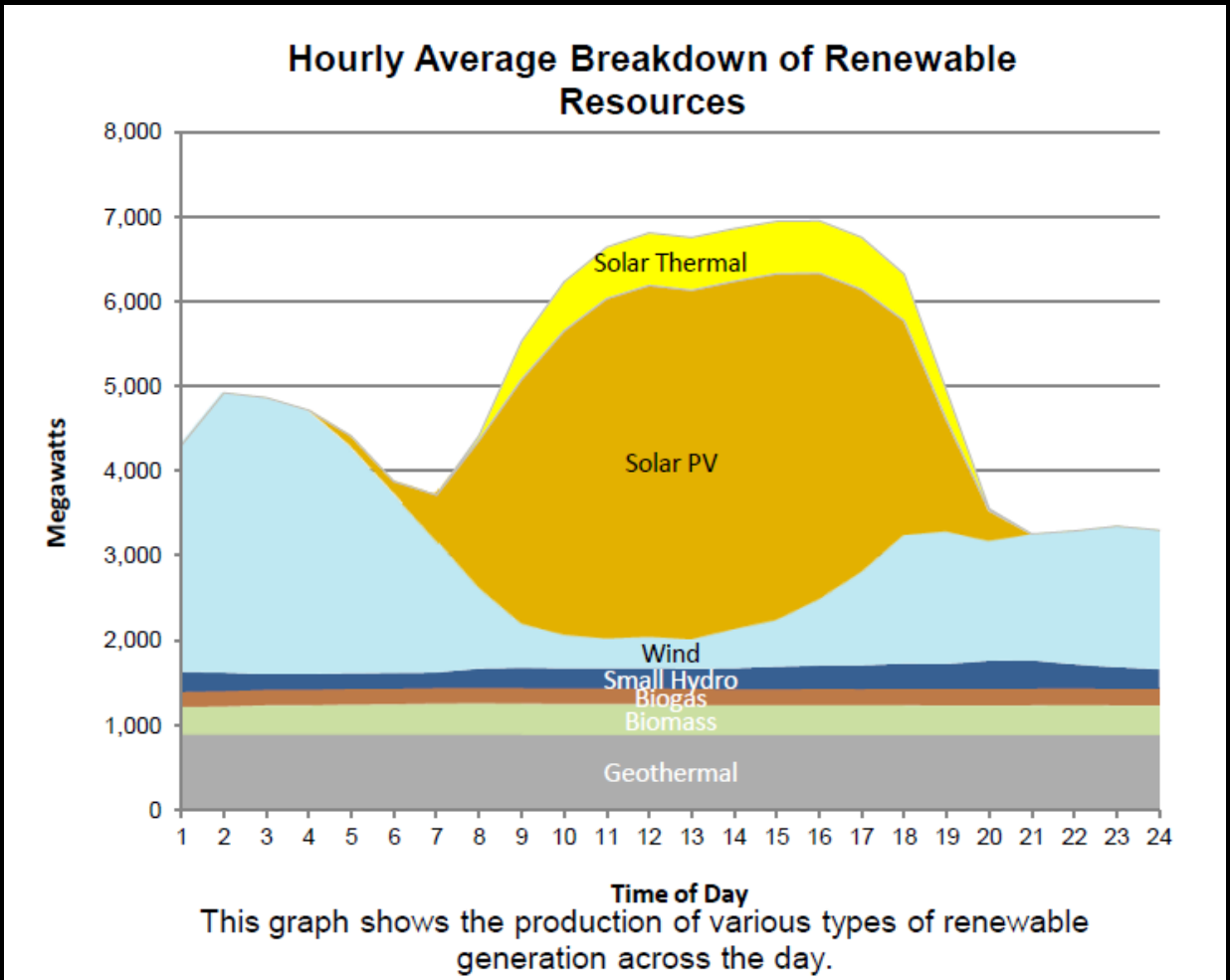


Over the past decade, the cost to produce renewable energy has fallen drastically as the technology has improved. While it varies by region, in many situations renewables now provide cheaper energy than fossil fuel alternatives. Unsurprisingly, this has spurred greater adoption. What makes renewables really appealing is that, once you have the infrastructure in place, they can generate electricity at low to no costs moving forward. Sunshine is free, coal is not. This characteristic is why you see mining farms concentrated in areas like China, which has cheap hydroelectric power, and Iceland, given its abundant geothermal energy. In China, you even see miners relocate their operations to different parts of the country during the rainy season. They do this to benefit from the excess (i.e. cheap) hydro power, provided by the increased rainfall, that would otherwise go to waste. We should note that coal, the dirtiest of all fuels, is the other main energy source that Bitcoin miners currently use in China but, as seen in the charts below, that is already starting to be priced out. Recent estimates suggest that anywhere from 39-76% of all the energy used in Bitcoin mining comes from renewables. **That's likely more than it was 5 years ago, and likely less than it will be 5 years from now.**



If we want to reverse and mitigate the impact of climate change, the world needs to invest in building out renewable energy infrastructure as soon as possible. The more investment renewables receive, the faster society can move away from dirty sources like fossil fuels. The best way to spur investment into renewable energy is to make it more profitable to operate. The problem that renewables often suffer from, particularly solar, is that they generate a lot of energy during their peak hours, but they have less consistent output throughout the entire day. For example, a solar farm would generate a lot of power around noon but none at night, while a plant that operates on fossil fuels can generate consistent energy every hour of the day.

For that solar farm, they likely generate so much energy in the middle of the day that they can't even use it all. The operators either need to store that energy for later (i.e. to provide electricity at night), or sell it back to the grid if there's demand for it, otherwise it goes to waste. This is partially why so much effort is being put towards improving battery technology, because if you can make storing that energy cheaper, renewables become much more profitable, and thus viable, to operate. While the cost of battery storage has also gotten significantly cheaper in recent years, it's still a work in progress. In addition, battery production comes with its own negative externalities that impact the environment (e.g. lithium mining). **Bitcoin offers an alternative solution, it's a digital battery.**



This graph shows the production of various types of renewable generation across the day.

Instead of letting that excess solar energy go to waste, it can be used to mine Bitcoin. **This alchemical process transmutes the electricity into liquid value. The solar farm operator can now monetize 100% of the energy being produced rather than a fraction of it.** The impact that has on the bottom line can be the difference between funding new solar infrastructure right now or waiting until the economics improve. Due to this**, Bitcoin mining can be an important catalyst for the green revolution because it subsidizes the production of renewable energy, filling the void left by many governments around the world.** This isn't just theory or speculation, this is how the mining industry already operates.

Does it matter how much 'total' energy Bitcoin uses as long as it comes from clean, renewable sources? Should we discourage Bitcoin mining even though it incentivizes more renewable energy to come online, not less? On both points I'd argue no. With that being said, coal and natural gas do account for their fair share of mining today. There's no arguing that. But all modern energy consumption is sourced from both clean and dirty sources. If governments want Bitcoin mining to be better for the environment, all they need to do is tax carbon emissions. In

most regions, a tax like this would immediately push miners towards renewable alternatives. **The cheapest energy source will always win this game.**

What's special about Bitcoin is that it provides a global incentive for nations to change their energy split from being 70% fossil fuels and 30% renewables, to 70% renewables and 30% fossil fuels. Ultimately, in the hunt to continuously improve profit margins, 100% renewable energy will likely be necessary. After all, sunshine is free but coal isn't.